

Framework - Building Sustainable Cities: Learning from Madrid's City Systems

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Final Assignment Title: *Building Sustainable Cities - Learning from Madrid's City Systems*

Topic: Sustainability

Subject Area: Global Studies / Environmental Education

Student's grade: 1 ESO

Unit Theme: Sustainability in Urban Systems

Lesson Overview: This series of lessons will introduce students to the concept of sustainability within city transportation systems, focusing on Madrid. Through engaging activities, critical discussions, and a final group project, students will learn how sustainable transportation impacts cities, the environment, and quality of life.

Lesson 1: Introduction to Urban Transportation Systems

Objective: Students will identify different urban transportation systems and discuss their environmental and social impacts.

Materials:

- Infographic: "Madrid Transportation at a Glance"
- Video: Overview of Madrid's Public Transport System (3-5 minutes)
- Worksheet: "Transportation Systems: Pros and Cons"

Activities:

- Warm-Up Discussion (10 min): "How do you get to school?"
 - Students brainstorm different modes of transportation and share their experiences.

- Presentation (15 min): Overview of Madrid's transportation system, including Metro, buses, biking, and walking.
- Group Activity (20 min): Students complete the worksheet, categorizing transportation modes as sustainable or unsustainable.
- Reflection (5 min): Discuss why sustainable transport matters to cities.

Assessment: Completion of the worksheet.

Lesson 2: Understanding Sustainability

Objective: Students will define sustainability and analyze how it applies to transportation systems. They will then start their brainstorm worksheet for their presentation.

Materials:

- Diagram: "The Three Pillars of Sustainability" (Environmental, Economic, Social)
- Short Article: "Forging new habits: how to make sustainable transport our new normal"
- Worksheet: BRAINSTORM

Activities:

- Mini-Lecture (10 min): Define sustainability and its three pillars with examples.
- Reading & Analysis (15 min): Students read the article and answer questions about its key points.
- Group Work (20 min): Students choose a city system and brainstorm the sustainability challenges associated. They will complete the BRAINSTORM worksheet.
- Debrief (5 min): Groups share their findings and discuss the challenges of achieving sustainability.

Assessment: Group analysis and BRAINSTORM worksheet completion.

Lesson 3: Designing a Sustainable Transportation System

Objective: Students will brainstorm and research their chosen city system for their project.

Materials:

- Worksheet: RESEARCH and PRESENTATION

Activities:

- Brainstorming Session (10 min): Review BRAINSTORM worksheet. What would your ideal transportation system look like?
- Design Challenge (30 min): In groups, students use the RESEARCH worksheet to design a sustainable system, considering factors like energy use, accessibility, and cost. They will research their chosen city system and start the PRESENTATION worksheet if finished.
- Gallery Walk (10 min): Groups present their designs to the class from the RESEARCH worksheet.

Assessment: Quality of group presentations and creativity in their designs.

Lesson 4: Final Project and Presentation

Objective: Students will synthesize their learning by creating a group presentation highlighting the city system of their choice - the history, problems, and solutions to become more sustainable.

Materials:

- Worksheet: PRESENTATION
- Rubric for Project Evaluation (see below)

Activities:

- Proposal Writing (20 min): Groups use their notes from previous lessons to write their proposals.
- Presentation Preparation (20 min): Groups create a visual aid presentation on the platform of their choice.
- Final Presentations (20 min): Groups present their presentations to the class.
- Feedback and Reflection (10 min): Peers and teacher provide constructive feedback on presentations.

Assessment: Based on RUBRIC.